

Final Submission Report

1. Project Overview

This repository presents an end-to-end MLOps workflow for heart disease prediction using the Cleveland heart disease dataset.

2. Dataset and EDA

The dataset is stored in `data/heart_disease.csv` and the repository includes a download helper at `scripts/download_data.py`.

3. Model Development

The training pipeline includes preprocessing, feature selection, model comparison, and serialization. The main training script is `train.py`.

4. Experiment Tracking

MLflow is used for experiment tracking. Training runs log parameters, metrics, and artifacts into `mlruns/`.

5. CI/CD and Testing

CI is implemented using GitHub Actions in `.github/workflows/ci.yml`. Tests are available in `tests/` and are executed automatically on pull requests.

6. Containerization and Deployment

The FastAPI service is containerized with Docker using `docker/Dockerfile` and `docker/docker-compose.yml`. A deployment script is provided in `scripts/deploy.py`.

7. Local Access Instructions

- API docs: `http://127.0.0.1:8000/docs`
- Streamlit UI: `http://127.0.0.1:8501`
- Run tests: `pytest -q`
- Run API: `uvicorn src.api.main:app --host 0.0.0.0 --port 8000`

8. Submission Assets

- Code and scripts: repository root
- Tests: `tests/`
- Screenshots: `screenshots/`
- Notebooks: `notebooks/`
- Final report: `docs/final_submission_report.md`